

Olist E-commerce Performance Analysis Report

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1 Project Overview

This project provides a comprehensive analysis of the Olist E-commerce dataset. The objective is to evaluate business performance through key metrics, including revenue generation, customer satisfaction, and logistical efficiency, providing actionable insights for decision-making.

2 Project Documentation

You can access the complete source code, Python data ingestion scripts, and project documentation on GitHub:

GitHub Repository: [Olist E-commerce Performance Analysis](https://github.com/komal-tiwari/olist-e-commerce-performance-analysis)

3 Dataset Information

The dataset used for this analysis is the **Brazilian E-Commerce Public Dataset** by Olist.

- **Source:** Kaggle (Olist Store)
- **Access Link:** <https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce>
- **Note:** Due to the large file size of the raw CSV datasets, the source files are not hosted in the GitHub repository. For reproducibility, the data can be downloaded directly from the Kaggle link provided above.

4 Dashboard Overview

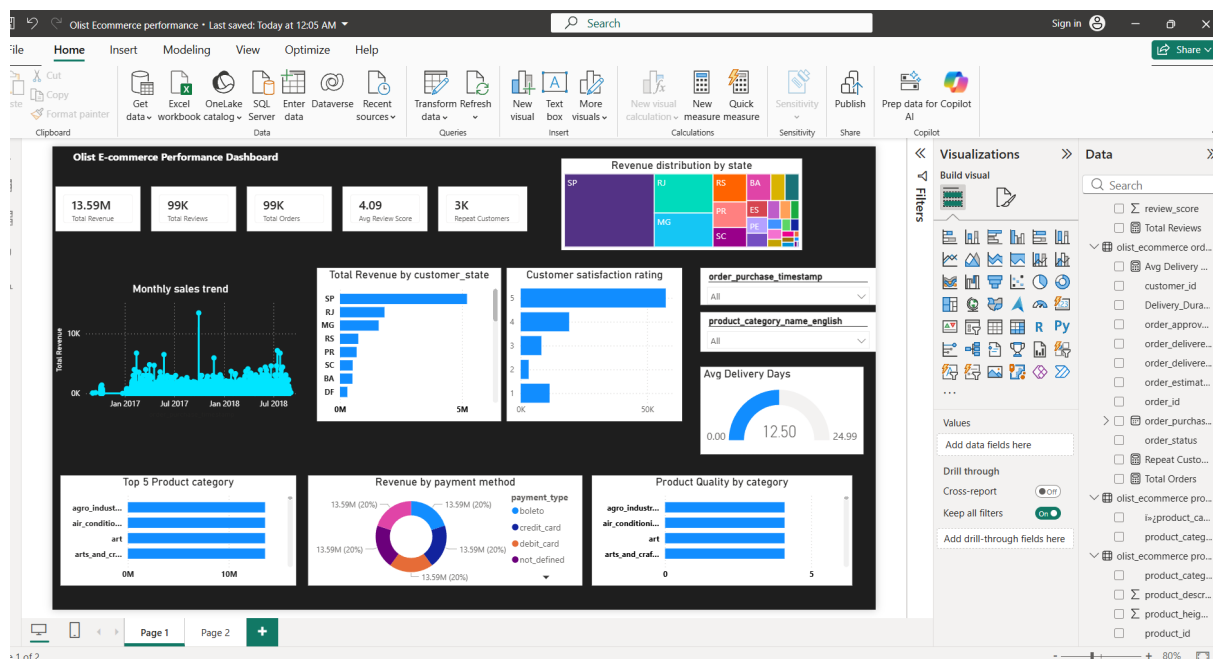


Figure 1: Olist E-commerce Performance Dashboard

5 Technical Stack

- **Data Manipulation:** Python (Pandas) for large-scale data ingestion and cleaning.
- **Data Querying:** SQL for relational database management and data aggregation.
- **Visualization:** Microsoft Power BI for interactive dashboarding.
- **Analytical Engine:** DAX (Data Analysis Expressions).

6 Methodology

6.1 Data Pipeline & Pre-processing

The data ingestion process utilized Python to handle high-volume files, ensuring memory optimization. We performed data cleaning to resolve inconsistencies in date formats and missing values. SQL was then used to perform joins, creating a unified dataset from separate order, payment, and customer tables.

6.2 Data Modeling

We implemented a **Star Schema** to facilitate effective data relationships. By creating a bridge between *olist_ecommerce_orders* and *olist_ecommerce_customers*, we ensured accurate tracking of customer identity versus individual transactions.

7 Analytical Logic (DAX)

To derive operational and business KPIs, we implemented custom DAX measures. Key measures developed include:

7.1 1. Average Delivery Days

This measure calculates the difference between order purchase and delivery, providing insights into logistics efficiency.

```
Avg Delivery Days =  
AVERAGEX(  
    'olist_ecommerce_orders',  
    DATEDIFF('olist_ecommerce_orders'[order_purchase_timestamp],  
             'olist_ecommerce_orders'[order_delivered_customer_date], DAY)  
)
```

7.2 2. Total Revenue

Aggregating the total payment value from the order payments table.

```
Total Revenue = SUM('olist_ecommerce_order_payments'[payment_value])
```

7.3 3. Repeat Customer Loyalty Metric

To identify loyal customers by filtering unique IDs with more than one order.

```
Repeat Customers =  
VAR CustomersWithMultipleOrders =  
    FILTER(  
        VALUES('olist_ecommerce_customers'[customer_unique_id]),  
        CALCULATE(COUNT('olist_ecommerce_orders'[order_id])) > 1  
    )
```

RETURN COUNTROWS (CustomersWithMultipleOrders)

8 Key Insights & Recommendations

1. **Geographical Strategy:** The state of 'SP' represents our highest revenue hub. We recommend focusing regional marketing efforts here.
2. **Operational Efficiency:** Current delivery times indicate a need for logistical optimization in underperforming states to improve the customer satisfaction score.
3. **Retention Strategy:** With an identified cohort of repeat customers, implementing a loyalty program could significantly enhance Customer Lifetime Value (CLV).